Perform the following operations using Python on the Heart Diseases data sets a. Data cleaning b. Data integration c. Data transformation d. Error correcting e. Data model building

In [1]:
```
# import pandas library
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score,confusion_matrix
from sklearn.linear_model import LogisticRegression
import seaborn as sns
import matplotlib.pyplot as plt
```

In [2]:
```
# Reading csv file
df = pd.read_csv("dataset_Heart.csv")
df.head()
```

Out[2]:

	age	sexe	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	th
0	64	1	0	128	263	0	1	105	1	0.2	1	1	
1	46	0	2	142	177	0	0	160	1	1.4	0	0	
2	65	0	2	140	417	1	0	157	0	0.8	2	1	
3	66	1	0	160	228	0	0	138	0	2.3	2	0	
4	57	1	0	165	289	1	0	124	0	1.0	1	3	

In [3]:
```
df = df.drop_duplicates()
```

In [4]:
```
# Count ,min,max ,etc of each column
df.describe()
```

Out[4]:

	age	sexe	cp	trestbps	chol	fbs	reste
count	212.000000	212.000000	212.000000	212.000000	212.000000	212.000000	212.000
mean	54.731132	0.679245	1.000000	132.033019	244.976415	0.160377	0.514
std	9.159377	0.467871	1.066509	17.370887	53.194665	0.367824	0.519
min	29.000000	0.000000	0.000000	94.000000	131.000000	0.000000	0.000
25%	48.000000	0.000000	0.000000	120.000000	209.000000	0.000000	0.000
50%	56.000000	1.000000	1.000000	130.000000	237.500000	0.000000	1.000
75%	62.000000	1.000000	2.000000	140.000000	271.750000	0.000000	1.000
max	76.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000

In [5]:
```
# Information about each column data
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 212 entries, 0 to 211
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       212 non-null    int64
 1   sexe      212 non-null    int64
 2   cp        212 non-null    int64
 3   trestbps  212 non-null    int64
 4   chol      212 non-null    int64
 5   fbs       212 non-null    int64
 6   restecg   212 non-null    int64
 7   thalach   212 non-null    int64
 8   exang     212 non-null    int64
 9   oldpeak   212 non-null    float64
 10  slope     212 non-null    int64
 11  ca        212 non-null    int64
 12  thal      212 non-null    int64
 13  target    212 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 23.3 KB
```

In [6]:

```
#Finding null values in each column
df.isna().sum()# Information about each column data
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 212 entries, 0 to 211
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       212 non-null    int64
 1   sexe      212 non-null    int64
 2   cp        212 non-null    int64
 3   trestbps  212 non-null    int64
 4   chol      212 non-null    int64
 5   fbs       212 non-null    int64
 6   restecg   212 non-null    int64
 7   thalach   212 non-null    int64
 8   exang     212 non-null    int64
 9   oldpeak   212 non-null    float64
 10  slope     212 non-null    int64
 11  ca        212 non-null    int64
 12  thal      212 non-null    int64
 13  target    212 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 23.3 KB
```

In [7]:

```
df.head()
```

Out[7]:

	age	sexe	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	th
0	64	1	0	128	263	0	1	105	1	0.2	1	1	
1	46	0	2	142	177	0	0	160	1	1.4	0	0	
2	65	0	2	140	417	1	0	157	0	0.8	2	1	
3	66	1	0	160	228	0	0	138	0	2.3	2	0	
4	57	1	0	165	289	1	0	124	0	1.0	1	3	

In [8]:
```
df.fbs.unique()
```

Out[8]:
```
array([0, 1])
```

In [9]:
```
subSet1 = df[['age','cp','chol','thalach']]
```

In [10]:
```
subSet2 = df[['exang','slope','target']]
```

In [11]:
```
merged_df = subSet1.merge(right=subSet2,how='cross')
merged_df.head()
```

Out[11]:

	age	cp	chol	thalach	exang	slope	target
0	64	0	263	105	1	1	1
1	64	0	263	105	1	0	1
2	64	0	263	105	0	2	1
3	64	0	263	105	0	2	1
4	64	0	263	105	0	1	0

In [12]:
```
df.columns
```

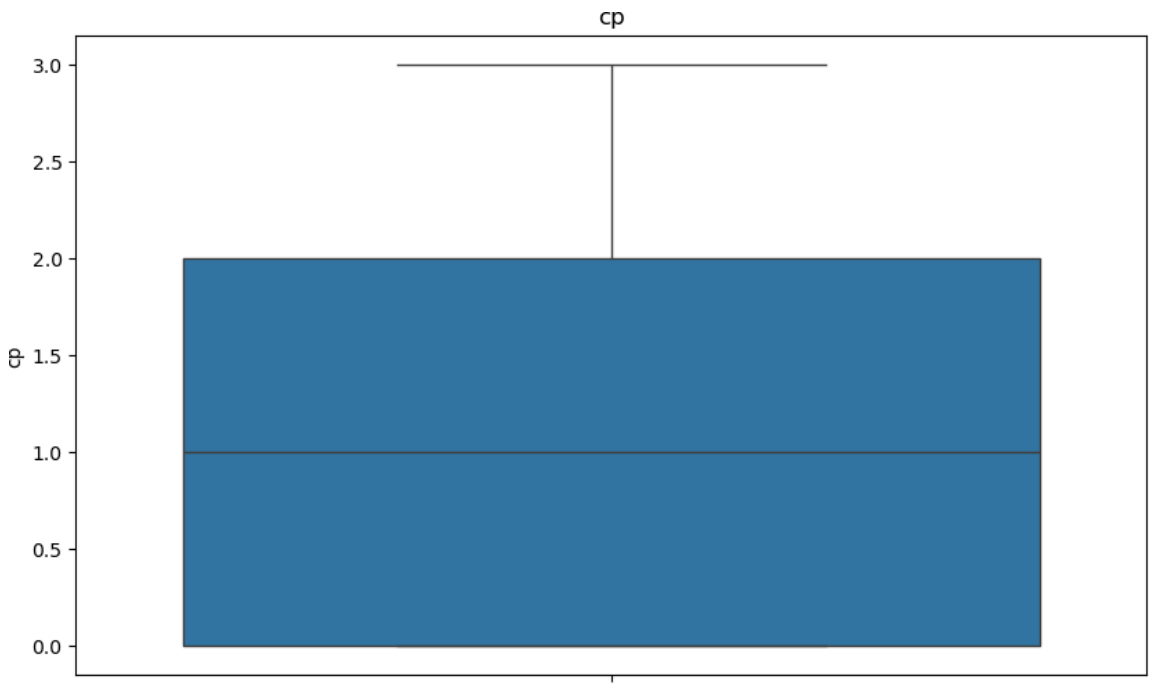
Out[12]:
```
Index(['age', 'sexe', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],
      dtype='object')
```

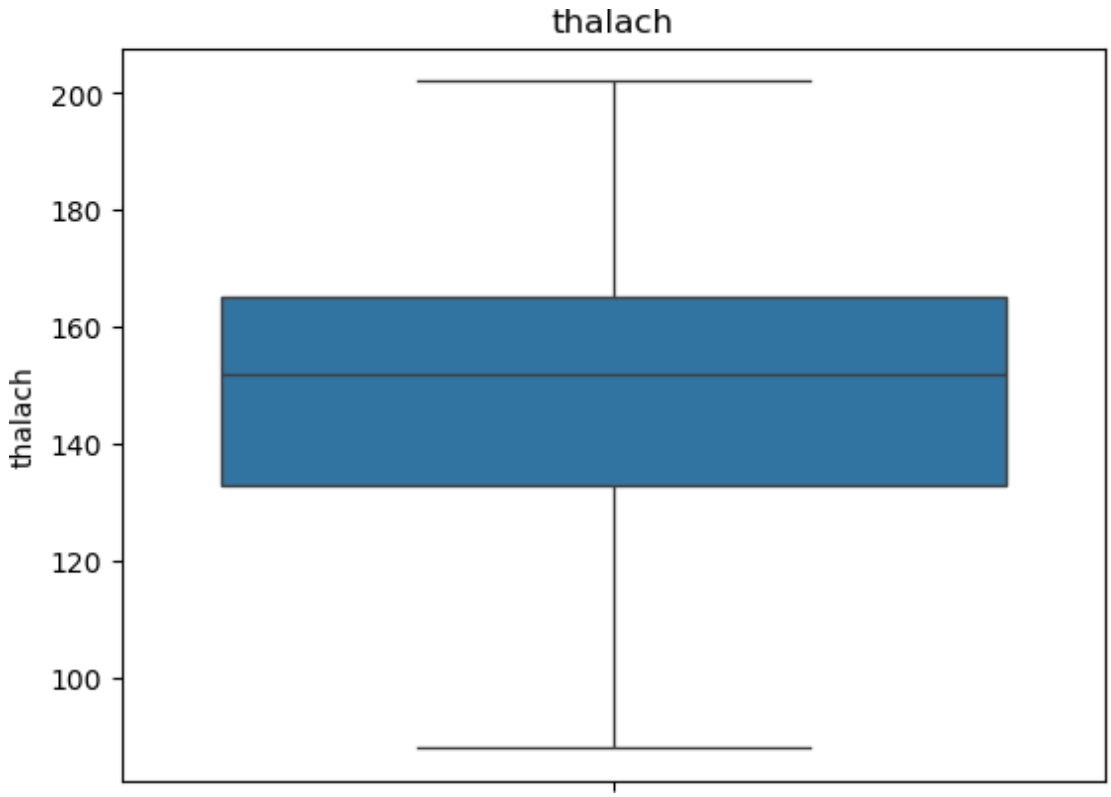
In [13]:
```
def remove_outliers(column):
    Q1 = column.quantile(0.25)
    Q3 = column.quantile(0.75)
    IQR = Q3 - Q1
    threshold = 1.5 * IQR
    outlier_mask = (column < Q1 - threshold) | (column > Q3 + threshold)
    return column[~outlier_mask]
```

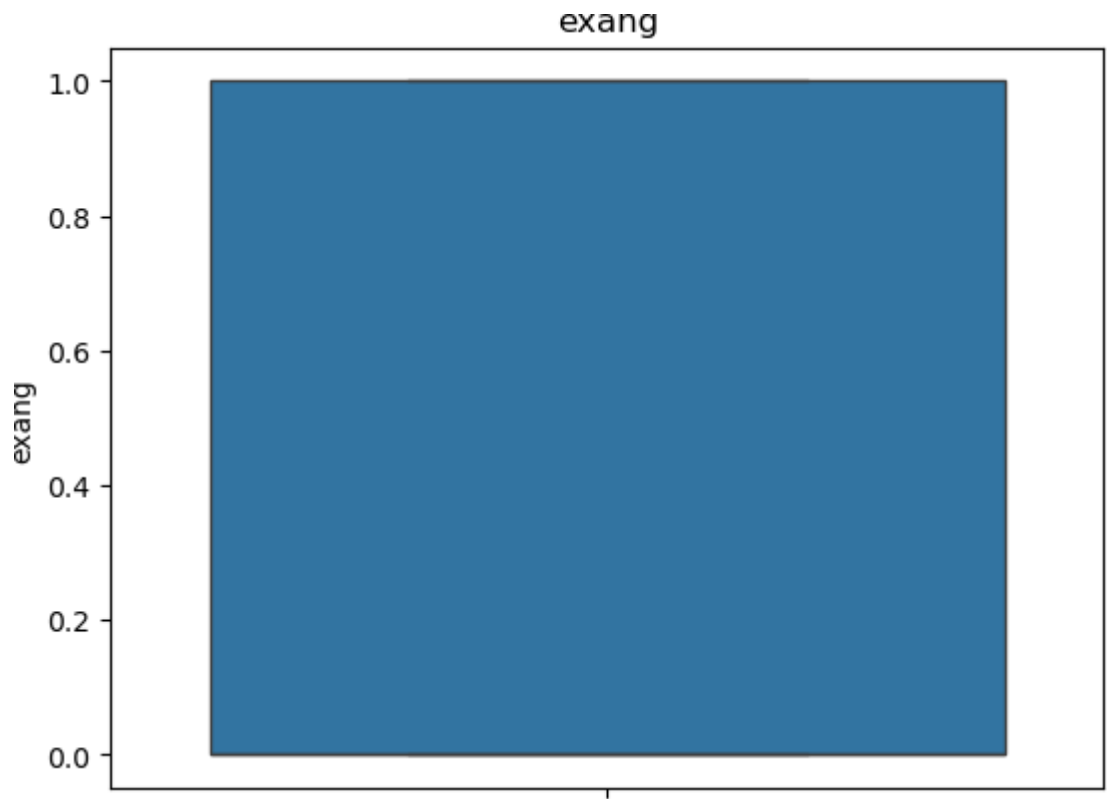
In [14]:
```
# Remove outliers for each column using a loop
col_name = ['cp','thalach','exang','oldpeak','slope','ca']
for col in col_name:
    df[col] = remove_outliers(df[col])
```

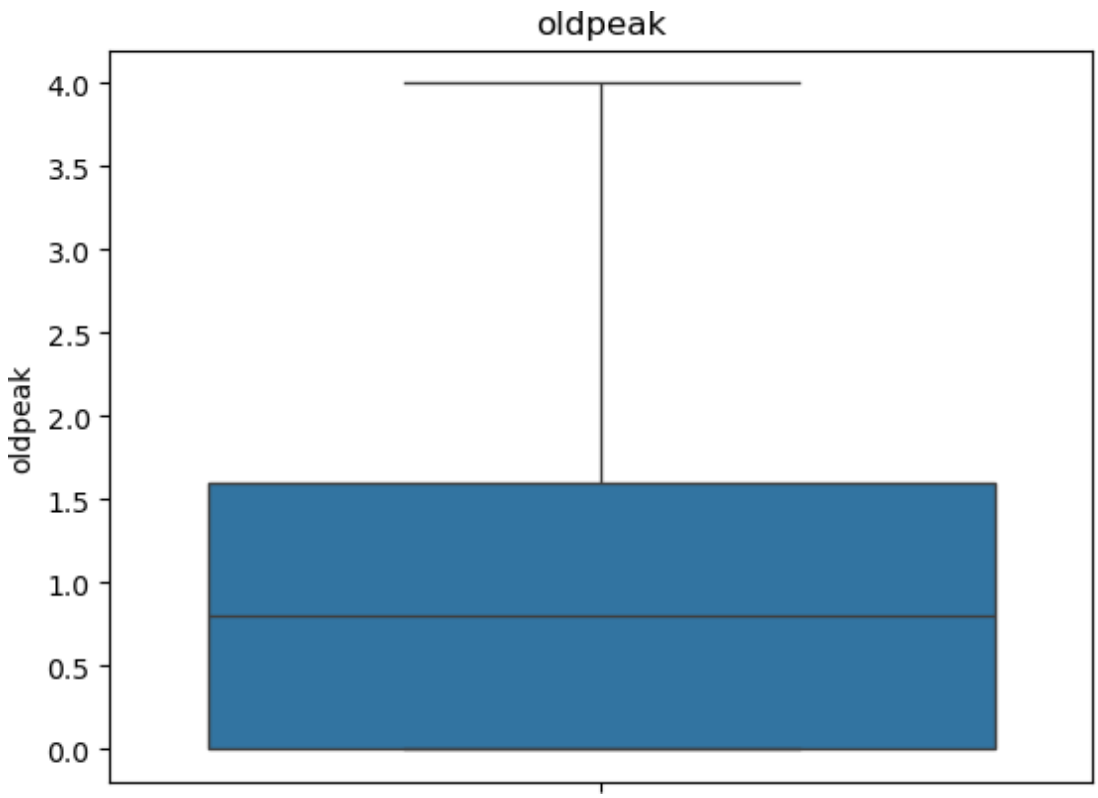
In [15]:
```
plt.figure(figsize=(10, 6))  # Adjust the figure size if needed
for col in col_name:
    sns.boxplot(data=df[col])
```

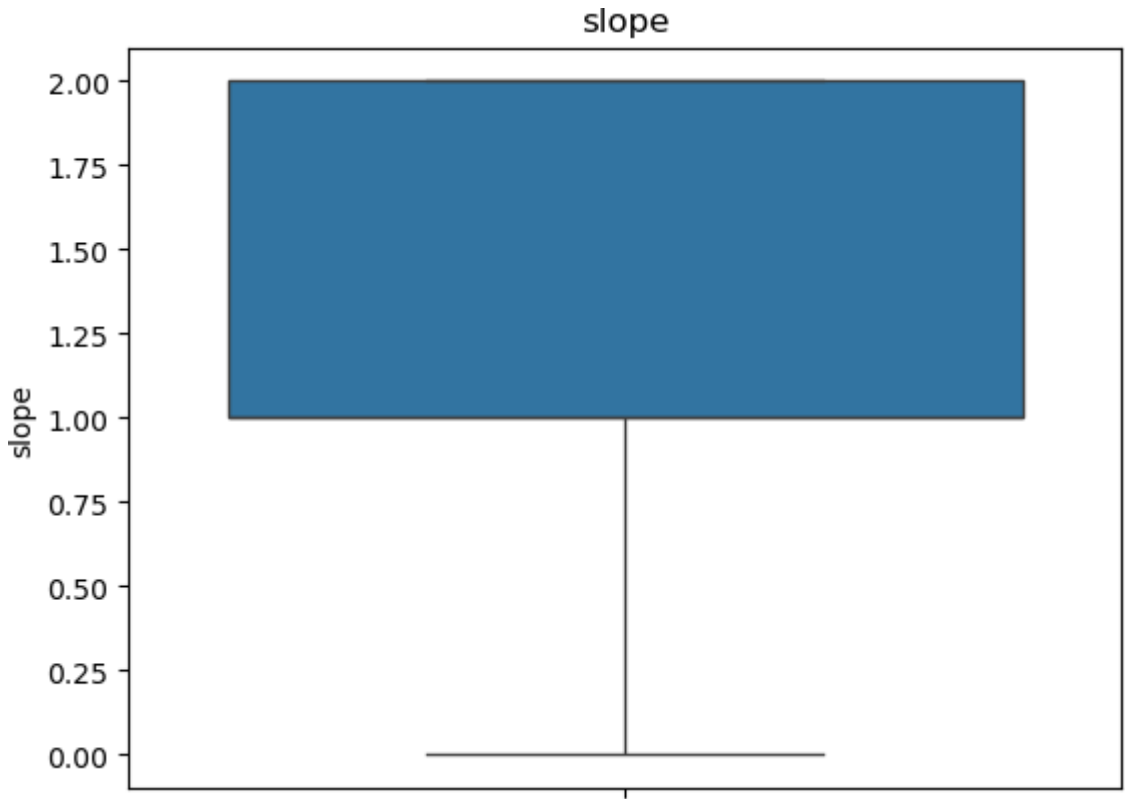
```
    plt.title(col)
    plt.show()
```

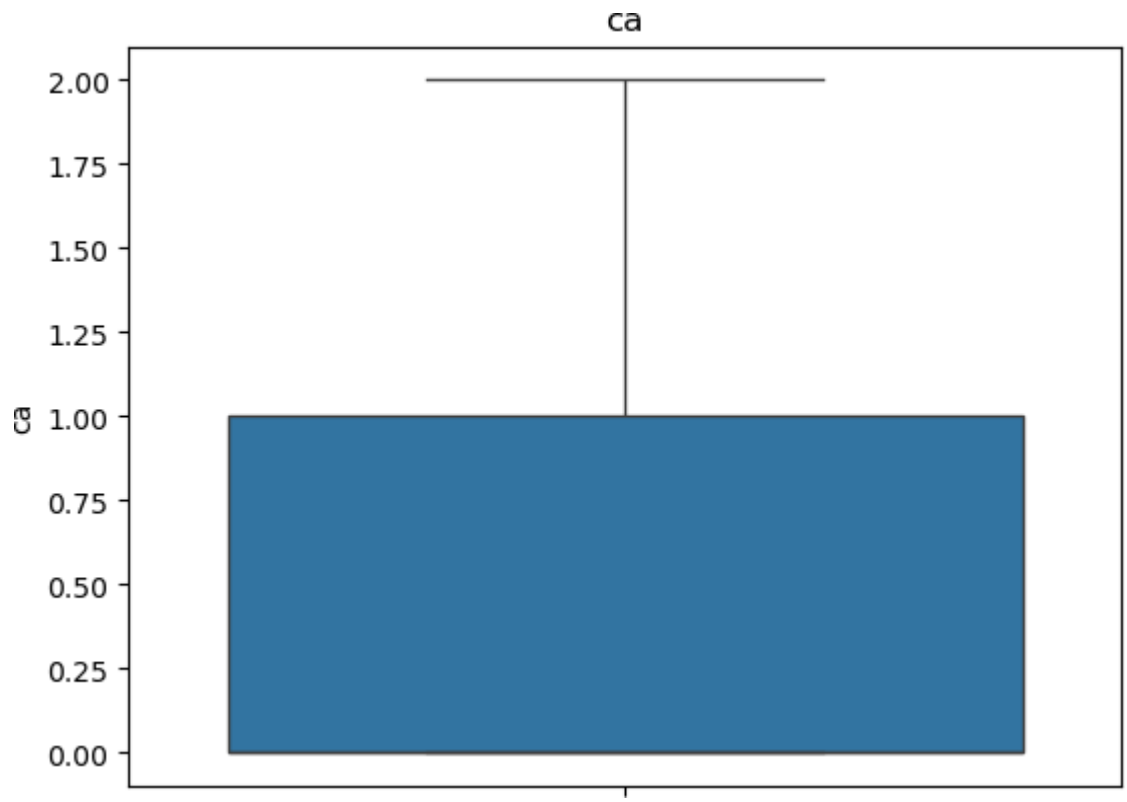





exang
exang
1.0
0.8
0.6
0.4
0.2
0.0


oldpeak
oldpeak
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0





In [16]:
```
df = df.dropna()
```

In [17]:
```
df.isna().sum()
```

Out[17]:
```
age         0
sexe        0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64
```

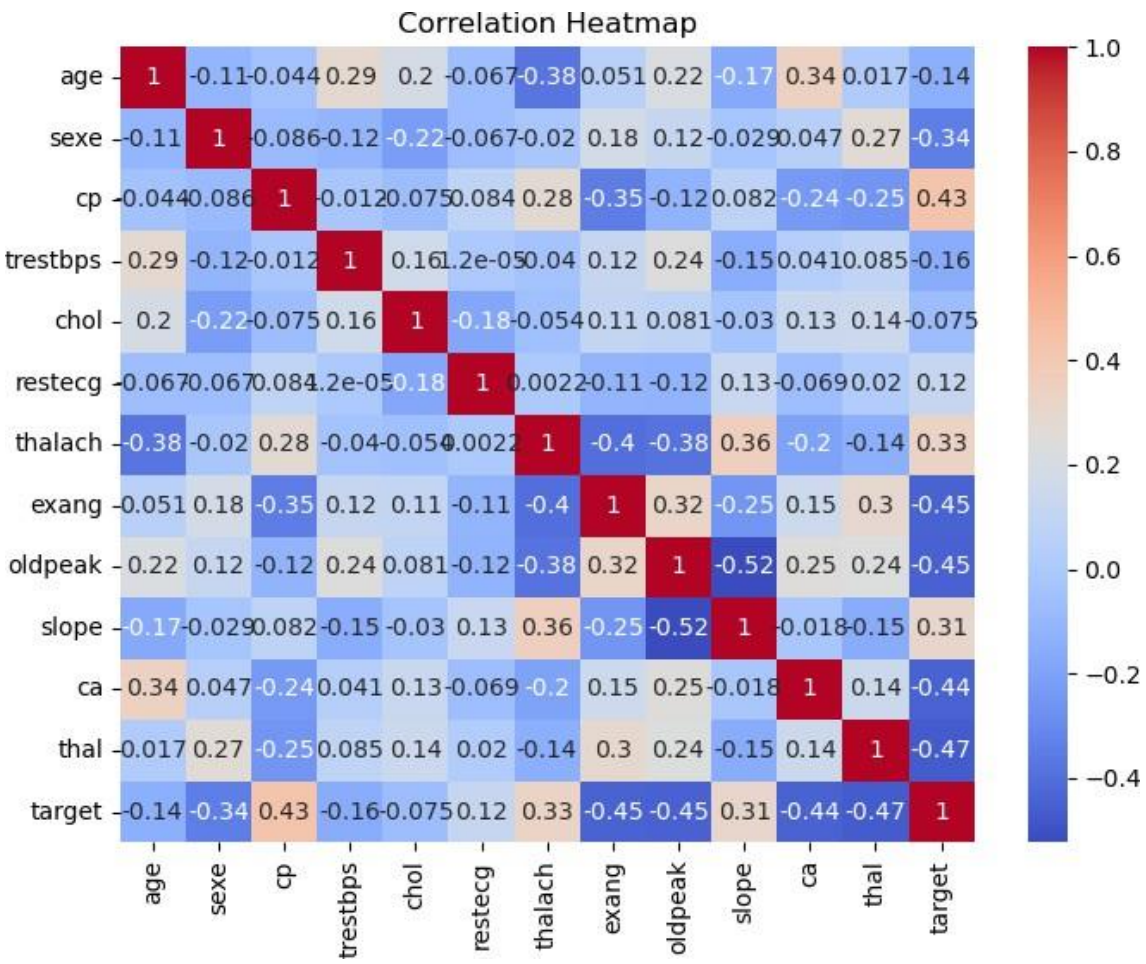
In [18]:
```
df = df.drop('fbs',axis=1)
```

In [19]:
```
# Compute correlations between features and target
correlations = df.corr()['target'].drop('target')

# Print correlations
print("Correlation with the Target:")
print(correlations)
print()

# Plot correlation heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

```
Correlation with the Target:
age         -0.141383
sexe        -0.342506
cp           0.432323
trestbps    -0.155796
chol        -0.074646
restecg      0.116095
thalach      0.328455
exang       -0.445435
oldpeak     -0.448892
slope        0.306024
ca          -0.441641
thal        -0.468544
Name: target, dtype: float64
```



In [20]:
```
# df.isna().sum()
```

In [22]:
```
# splitting data using train test split
x = df[['cp','thalach','exang','oldpeak','slope','ca']]
y = df.target
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=0)

x_train.shape,x_test.shape,y_train.shape,y_test.shape
```

Out[22]:
```
((153, 6), (39, 6), (153,), (39,))
```

In [23]:
```
from sklearn.preprocessing import StandardScaler
```

In [24]:
```
scaler = StandardScaler()
```

In [25]:
```
x_train_scaled = scaler.fit_transform(x_train)
x_test_scaled = scaler.transform(x_test)
```

In [26]:
```
y_train= np.array(y_train).reshape(-1, 1)
y_test= np.array(y_test).reshape(-1, 1)
```

In [27]:
```
y_train.shape
```

Out[27]:
```
(153, 1)
```

In [28]:
```
model = LogisticRegression()
model.fit(x_train_scaled, y_train)
```

```
# Make predictions on the test set
y_pred = model.predict(x_test_scaled)

# Evaluate the model's accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.8974358974358975
```

```
C:\Users\Rohit\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1406:   Data
ConversionWarning: A column-vector y was passed when a 1d array was expected. Ple
ase change the shape of y to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

In [29]:

```
#Classification model using Decision Tree
from sklearn.tree import DecisionTreeClassifier
tc=DecisionTreeClassifier(criterion='entropy')
tc.fit(x_train_scaled,y_train)
y_pred=tc.predict(x_test_scaled)

print("Training Accuracy Score :",accuracy_score(y_pred,y_test))
print("Training Confusion Matrix  :",confusion_matrix(y_pred,y_test))
```

```
Training Accuracy Score : 0.717948717948718
Training Confusion Matrix  : [[12  7]
 [ 4 16]]
```

In [ ]: